

Aly Ashr

Data Engineer

✉ alyashr33@gmail.com 📍 Monroe, NY, USA

PROFILE

Results-driven **Lead Data Engineer** with over **7 years** of expertise in architecting and deploying **scalable cloud-based data platforms** across diverse industries such as **healthcare, finance, e-commerce, and telecommunications**. Adept at building robust **real-time data pipelines** and implementing **big data technologies** such as **Apache Spark, Apache Kafka, Flink, and Hadoop** for high-performance data processing. Proven expertise in designing **ETL/ELT workflows** using **Apache Airflow, dbt, AWS Glue, Azure Data Factory, and Informatica** to streamline data integration across multiple systems. Strong background in **cloud data engineering** on platforms like **AWS, Azure, and GCP**, leveraging **Snowflake, Delta Lake, Redshift, and Google BigQuery** for data lakehouse architectures and storage. Experienced in implementing **data governance and compliance frameworks** such as **GDPR, CCPA, HIPAA, and ISO 27001**, ensuring data security and privacy across all systems. Skilled in leading teams in deploying **machine learning models** using **AWS SageMaker, Azure ML, Google AI Platform**, and integrating them into production environments for real-time analytics. Capable of optimizing **CI/CD pipelines** using **Jenkins, GitLab CI, and GitHub Actions**, alongside **Kubernetes and Docker** for seamless deployment and containerization of data applications. Strong communicator with the ability to work cross-functionally to deliver **data-driven solutions** that drive business outcomes, enhance operational efficiency, and reduce costs.

SKILLS

Big Data Processing:

Apache Spark, Hadoop, Apache Kafka, Kafka Streams, **Flink, Apache Flume, Apache Beam, AWS Kinesis, Google Dataflow**

ETL/ELT:

Apache Airflow, dbt, **AWS Glue, Talend, Informatica, Matillion, Apache NiFi, Azure Data Factory, DataBricks, Prefect**

Data Lakehouse & Storage:

Snowflake, Delta Lake, Redshift, S3, Google Cloud Storage, Azure Data Lake, AWS Lake Formation, Hadoop HDFS

Data Modeling:

Star Schema, Snowflake Schema, Data Vault, Medallion Architecture, Dimensional Modeling, OLAP Cubes, Data Warehouse Design

Real-Time Streaming:

Apache Kafka, AWS Kinesis, Apache Pulsar, Google Cloud Pub/Sub, Apache Flink, Azure Event Hubs, Kafka Streams

Cloud Platforms & Services:

AWS (Redshift, Lambda, S3, EMR, Athena, Glue, **SageMaker, AWS Lambda, AWS Batch, AWS Step Functions**), Azure (Data Factory, **Synapse Analytics, Databricks, Azure ML, Azure Kubernetes Service (AKS), Azure Blob Storage**), GCP (BigQuery, **Google Cloud Pub/Sub, Google Cloud Composer, Dataflow, Cloud Functions**)

DevOps & CI/CD:

Docker, **Kubernetes, ECS, AKS, Helm, Terraform, CloudFormation, ARM, Jenkins, GitLab CI, GitHub Actions, CircleCI, Ansible, GitOps**

Programming & Scripting:

Python, SQL, Java, Scala, Bash, Shell Scripting, **R, Go, JavaScript, TypeScript, SQL Server T-SQL**

Frameworks/Libraries:

PySpark, Pandas, NumPy, TensorFlow, PyTorch, scikit-learn, Keras, XGBoost, LightGBM, Dask, H2O.ai 🌐

ML & Model Deployment:

MLFlow, TensorFlow, **Kubeflow, Seldon Core, Scikit-learn, Keras, Triton Inference Server, AWS Sagemaker, Azure ML**

Cloud ML:

AWS SageMaker, Azure Machine Learning, Google AI Platform, Google AutoML, Amazon Rekognition, AWS Comprehend, Databricks ML

Business Intelligence & Data Visualization:

Power BI, Tableau, Looker, QlikView, Superset, Data Storytelling, Dashboards, **KPI Metrics, Geospatial Visualization, Predictive Analytics, Interactive Reports, Data Reporting**

Data Governance & Compliance:

GDPR, CCPA, HIPAA, SOC2, ISO 27001, Data Quality, IAM, Data Masking, Encryption (AES, SSL/TLS), Data Privacy, Data Lineage, Data Auditing

PROFESSIONAL EXPERIENCE

Benchspire Technologies

01/2023 – Present

Lead Data Engineer

- Spearheaded the design and implementation of **real-time data pipelines** using **Apache Kafka** and **Apache Spark**, processing over 10TB of data daily for a global e-commerce client.
- Architected a **data lakehouse** leveraging **Snowflake** and **Delta Lake**, optimizing both batch and real-time data processing workflows.
- Led the migration of on-premises data solutions to **AWS cloud-native architectures**, reducing data processing time by 30% and operational costs by 25%.
- Designed and optimized end-to-end **ETL processes** with **Apache Airflow**, incorporating data quality validation steps that reduced errors by 20%.
- Managed and mentored a team of 5 engineers, ensuring best practices in **data engineering**, **cloud infrastructure**, and **data security** were followed.
- Partnered with the **Data Science** team to deploy predictive models into production using **AWS SageMaker**, enabling real-time analytics on customer behavior.

DataSap

07/2020 – 12/2022

Senior Data Engineer

- Led the transformation of a legacy data warehouse to **AWS Redshift**, optimizing queries and improving performance by over 40%.
- Developed and maintained **ETL pipelines** using **dbt** and **Apache Airflow**, handling large-scale data from multiple sources with increased reliability.
- Integrated **real-time data streaming** pipelines with **Apache Kafka**, enabling rapid, event-driven insights for marketing and operations teams.
- Automated data quality checks, resulting in a 25% reduction in manual intervention for quality assurance and reporting.
- Delivered **interactive dashboards** using **Power BI** to executive teams, providing key performance insights and operational metrics.

KPMG

04/2018 – 06/2020

Data Engineer

- Designed batch **ETL pipelines** using **Apache Spark** and **Hadoop**, processing large data volumes from multiple systems for financial analytics.
- Built **data pipelines** with **Azure Data Factory**, integrating data from diverse cloud-based and on-premises systems into a unified storage platform.
- Assisted in the deployment of **ML models** into production using **MLFlow**, helping the business unlock actionable insights from historical data.
- Maintained high-performing **databases** including **MySQL** and **PostgreSQL**, ensuring the stability and security of the company's critical systems.

AlphaBridge

01/2016 – 03/2018

Associate Data Engineer

- Assisted in the development and maintenance of **ETL pipelines** for processing key business data, ensuring smooth integration across multiple platforms.
- Supported the migration of datasets from **legacy data systems** to cloud environments such as **AWS** and **Azure**.
- Optimized **SQL queries** to enhance data extraction performance, contributing to faster decision-making processes for business teams.

PROJECTS

1. Real-Time Data Pipeline for E-Commerce Platform

- **Technologies:** Apache Kafka, Apache Spark, AWS Kinesis
- Designed and implemented **real-time data pipelines**, processing user data and enabling personalized marketing campaigns for a leading e-commerce company.
- Leveraged **AWS Kinesis** and **Kafka** for low-latency, high-throughput data ingestion and analysis.

2. Data Lakehouse Architecture for Financial Services

- **Technologies:** Snowflake, Delta Lake, AWS S3
- Led the design and deployment of a **data lakehouse**, consolidating transactional and analytical data to provide real-time financial insights.
- Reduced operational costs by 40% by optimizing data storage and access workflows using **Delta Lake**.

3. Healthcare Data Integration & ML Model Deployment

- **Technologies:** Apache Airflow, MLFlow, AWS Sagemaker, Python
- Built an automated data pipeline to integrate **healthcare** data from multiple sources and deployed predictive models to forecast patient readmissions.
- Implemented **AWS Sagemaker** for **ML model deployment**, improving care coordination and resource allocation in the hospital.

4. Data Quality Monitoring System for E-Commerce

- **Technologies:** dbt, Apache Airflow, Python
- Developed an automated **data quality monitoring system**, ensuring high data integrity in critical business processes for an e-commerce client.

- Improved data accuracy by 25% and reduced operational errors by automating data validation checks and anomaly detection.

EDUCATION

Bachelor's Degree